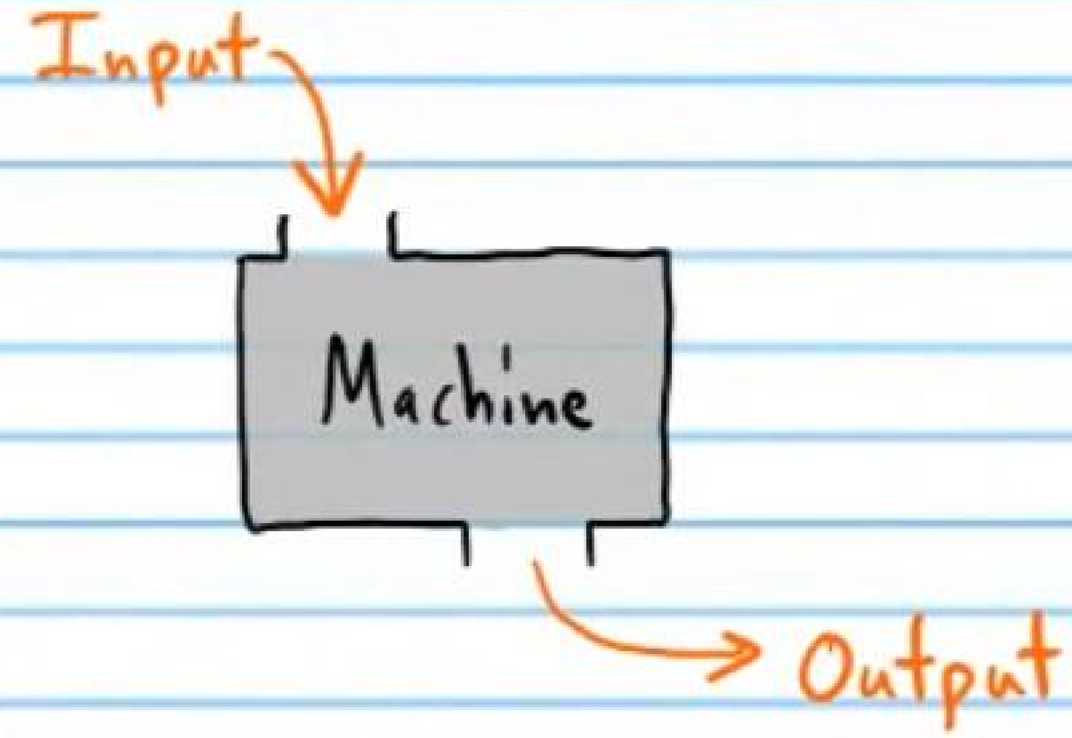
# Functions and Function Notation

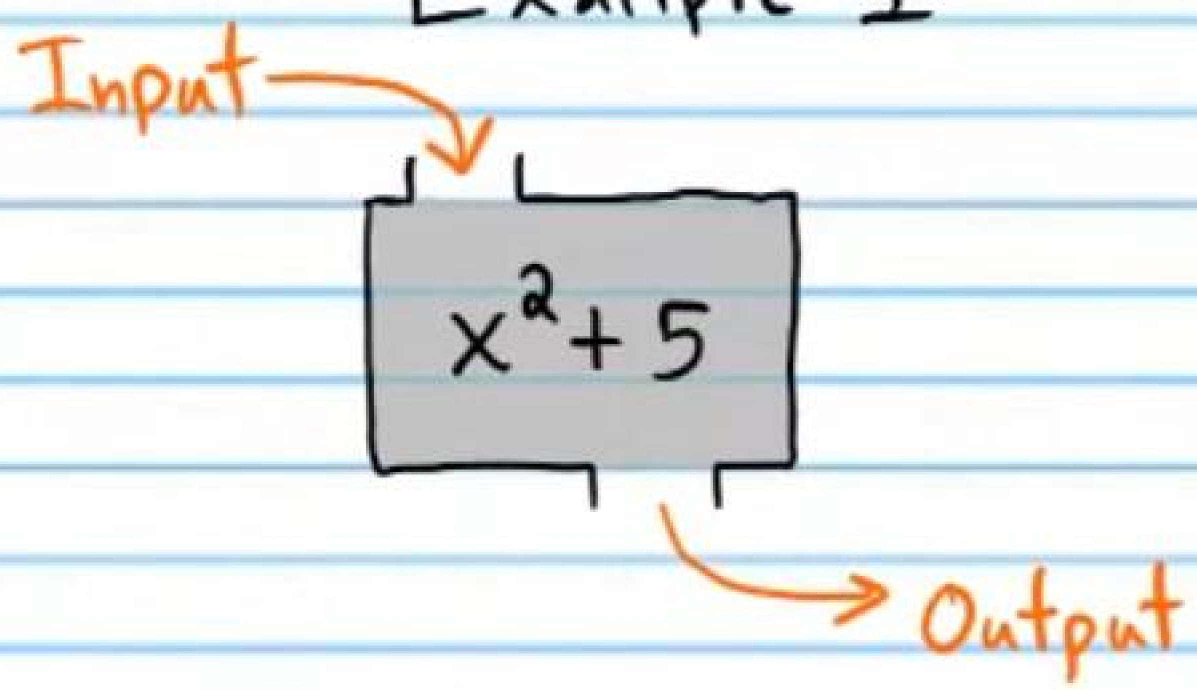
## The General Concept of a Function

A function can be thought of as a machine.

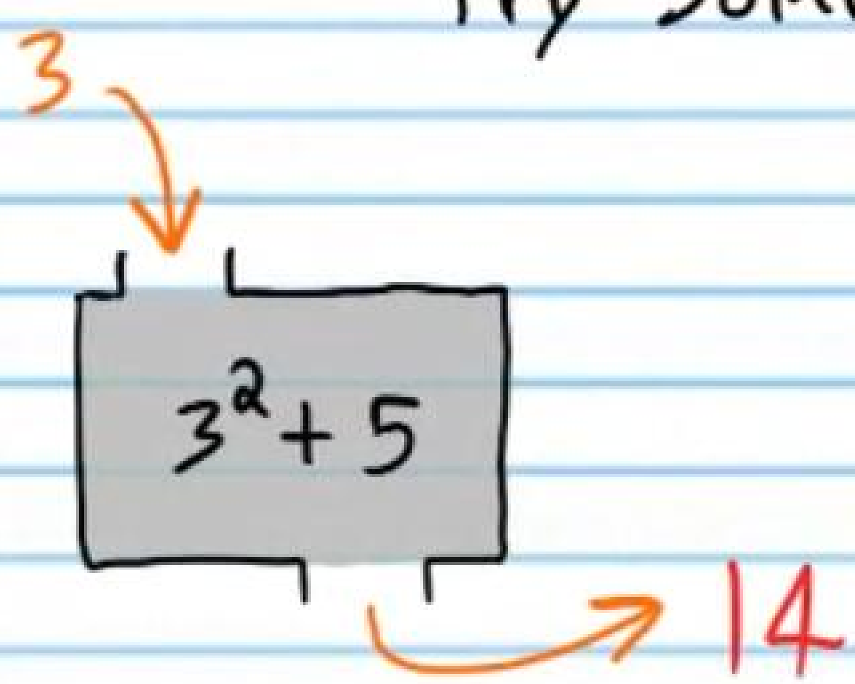


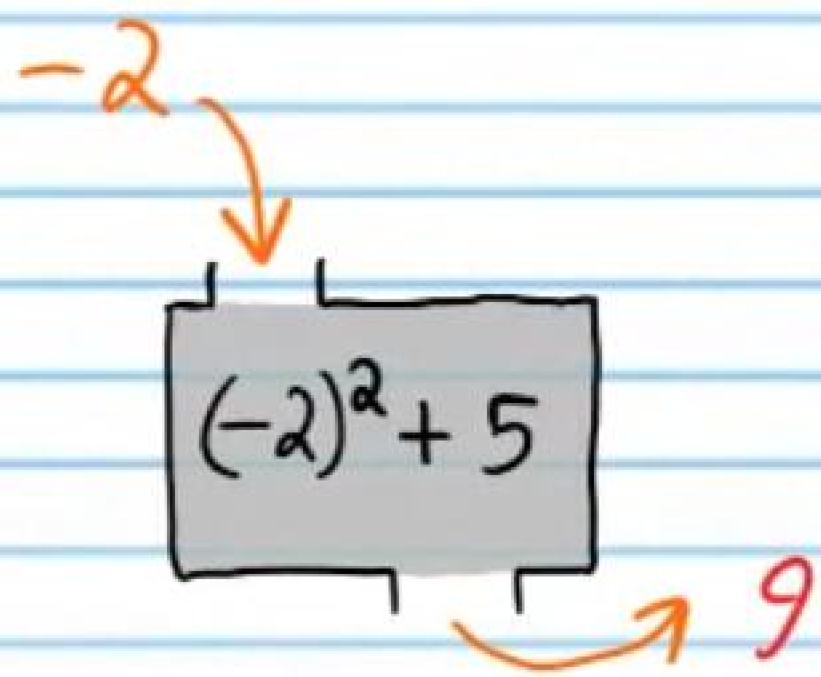

In the subject of mathematics, a function is a set of operations that are performed on a number.

### Example I

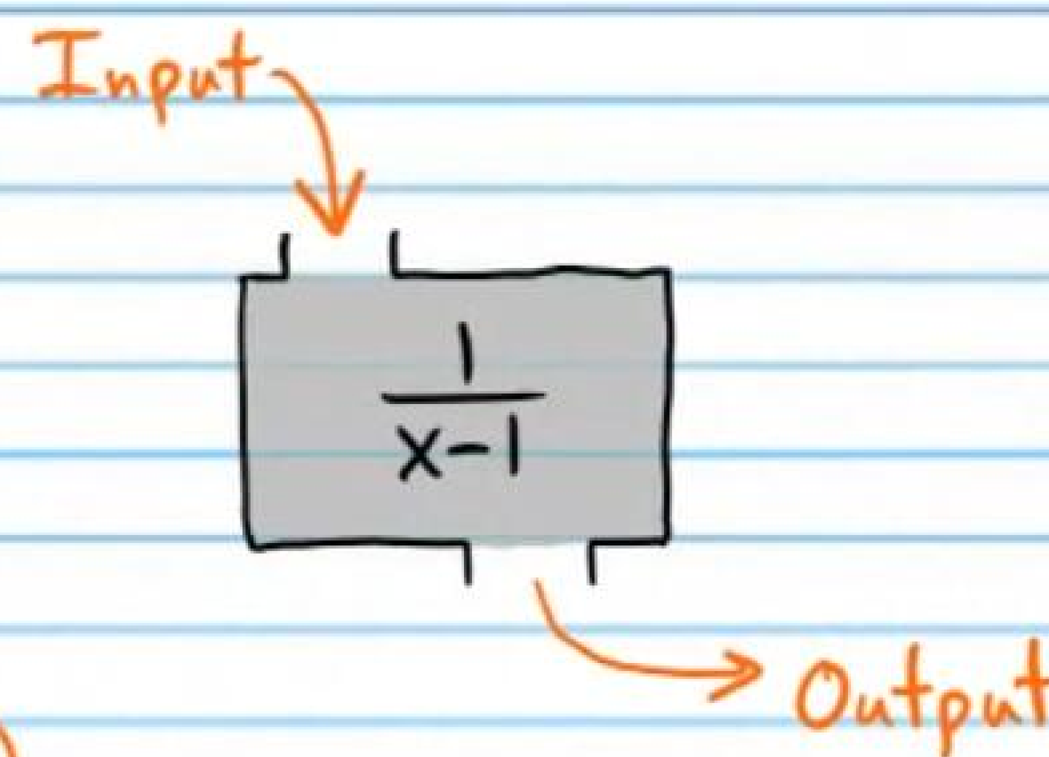


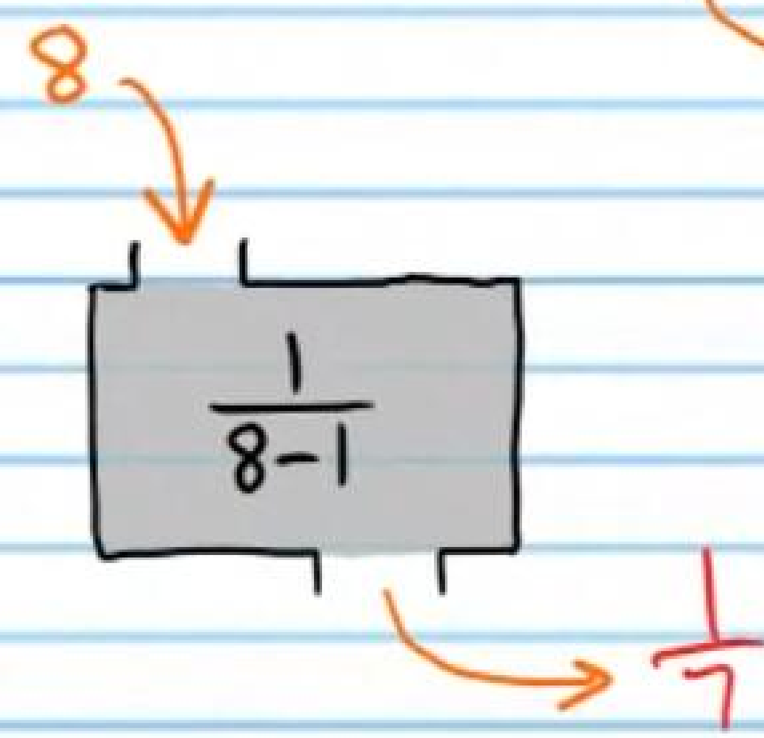

Try Some Inputs

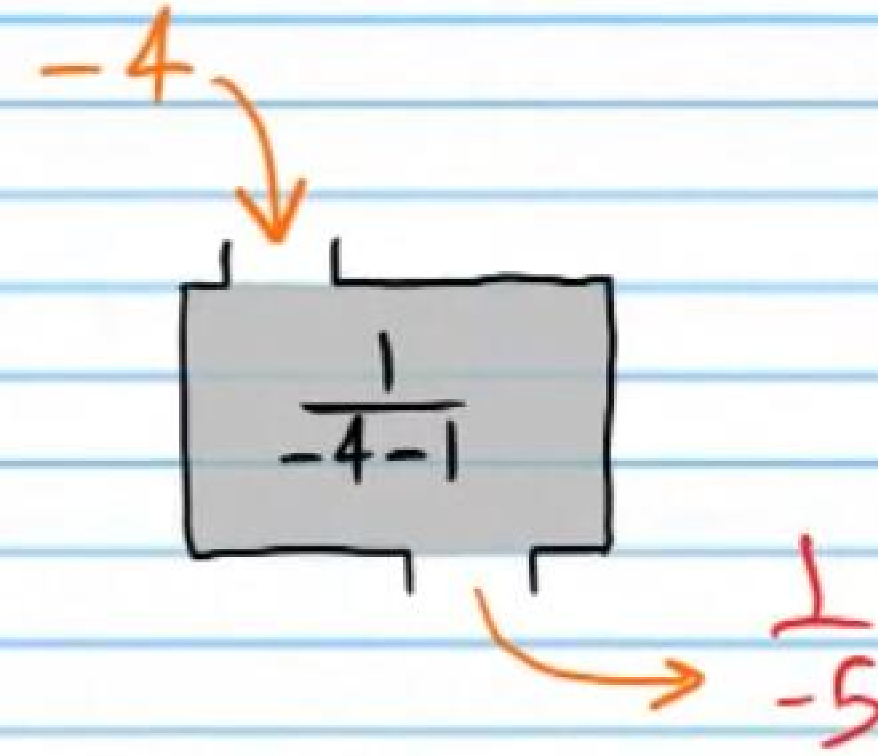


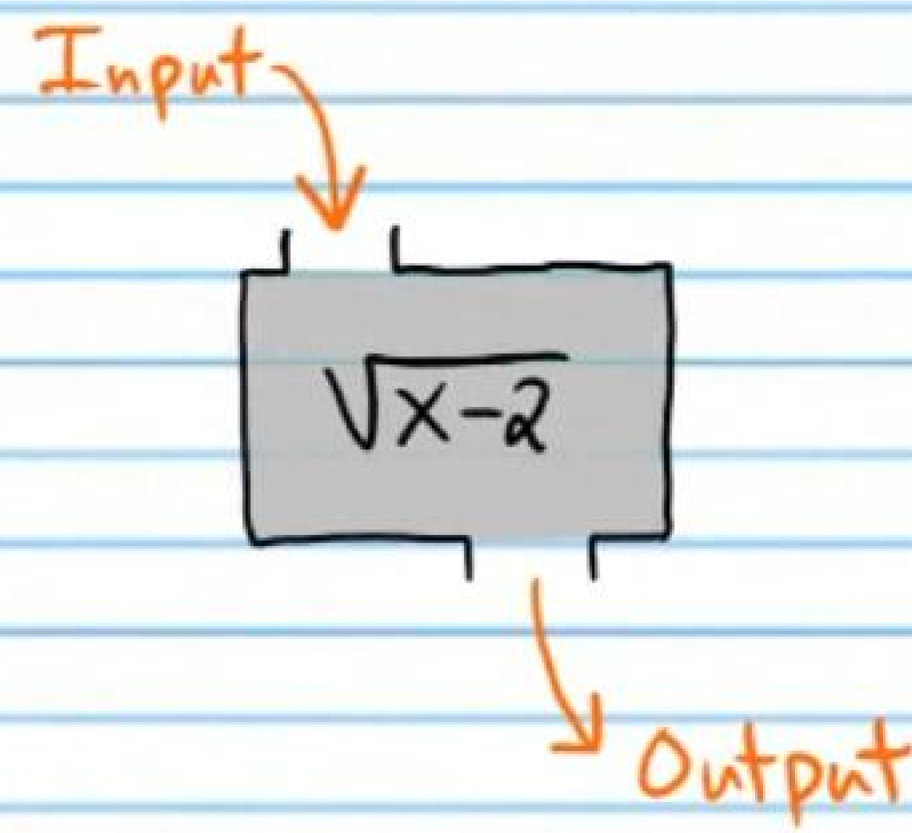

### Example II

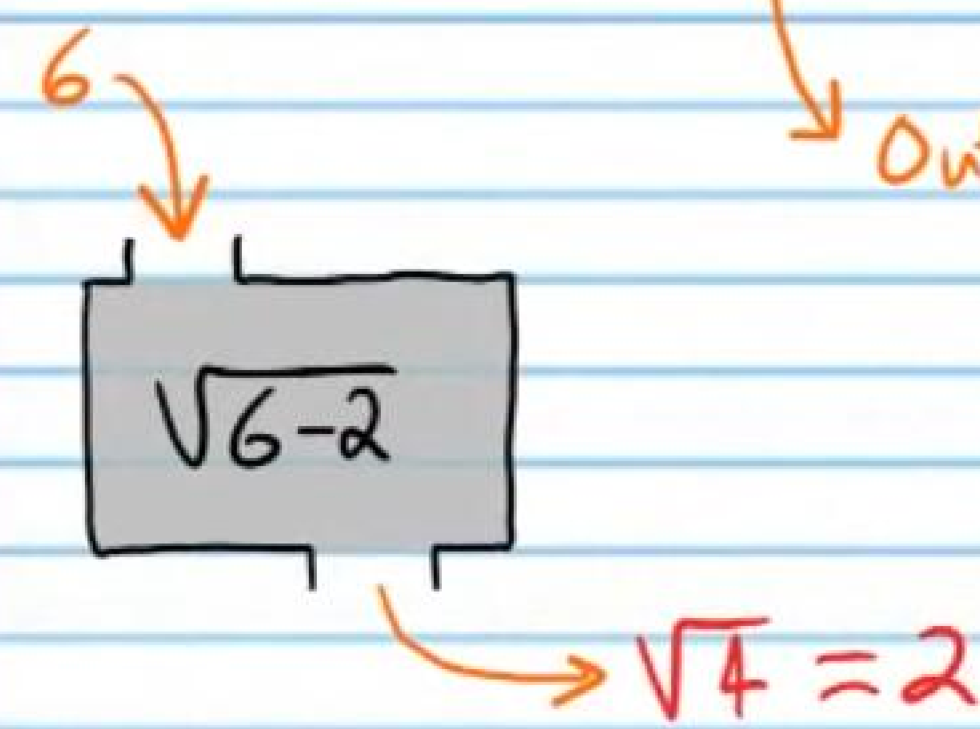


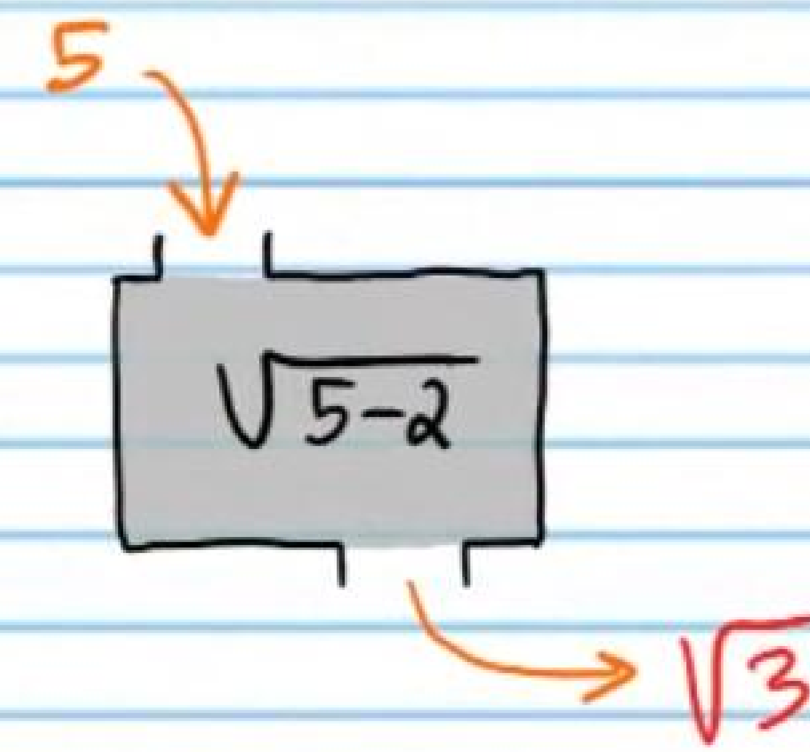



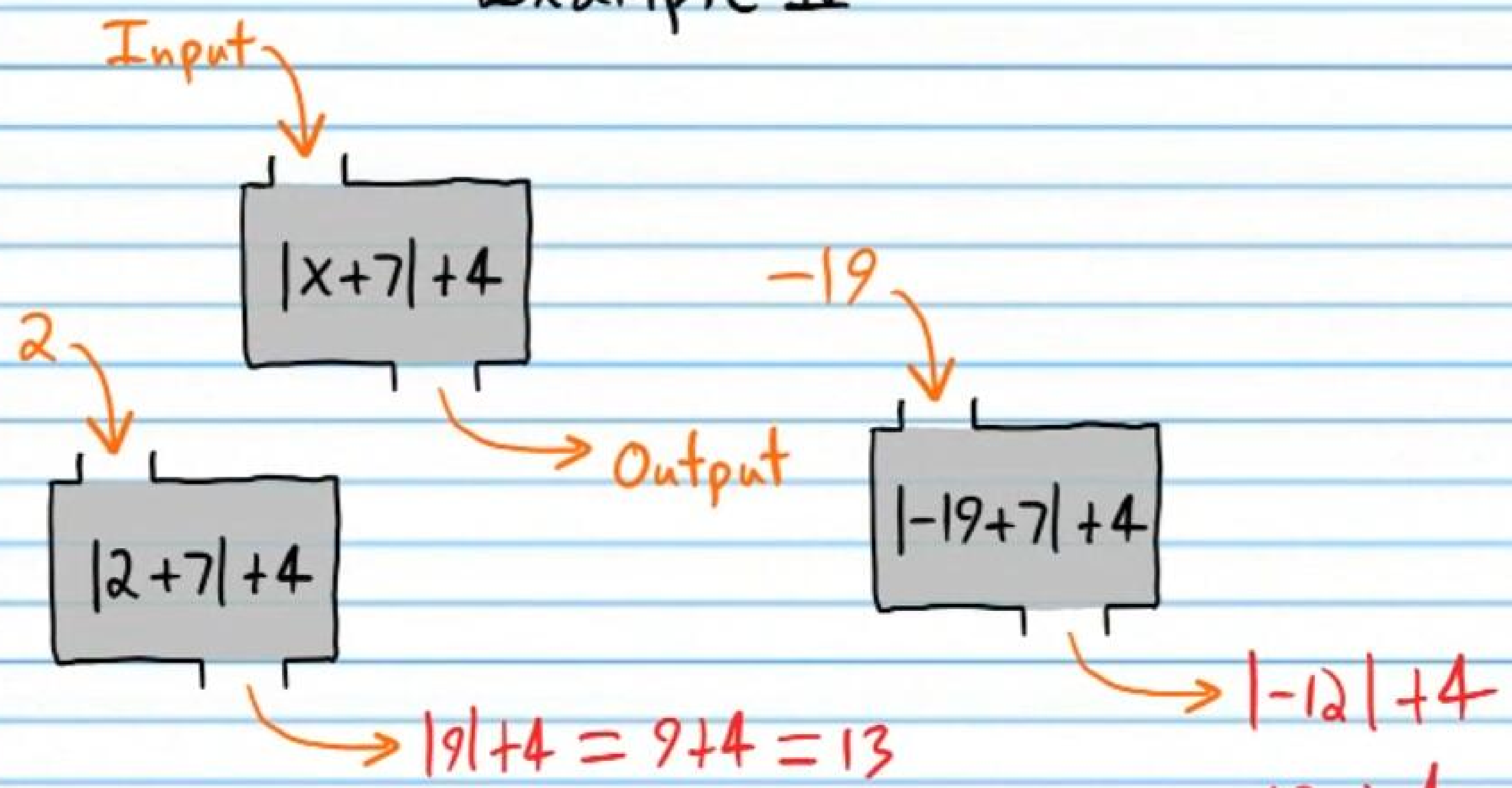

### Example III



## Example IV

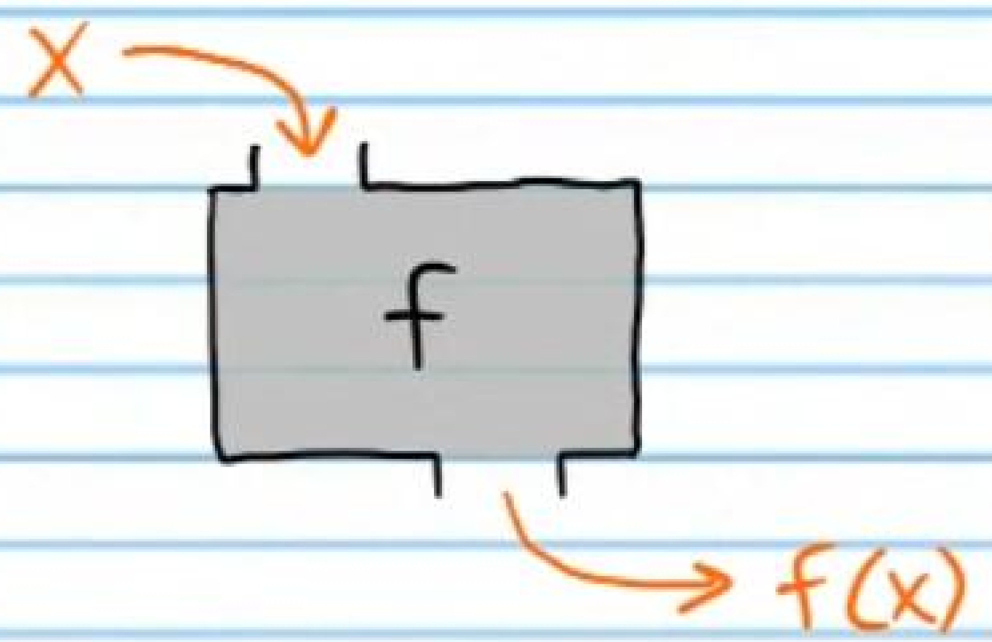



12 + 4

16

## Function Notation

"f" is the most common letter used for function notation, "x" is the most common letter used for inputs, and f(x) represents an output resulting from an input "x."

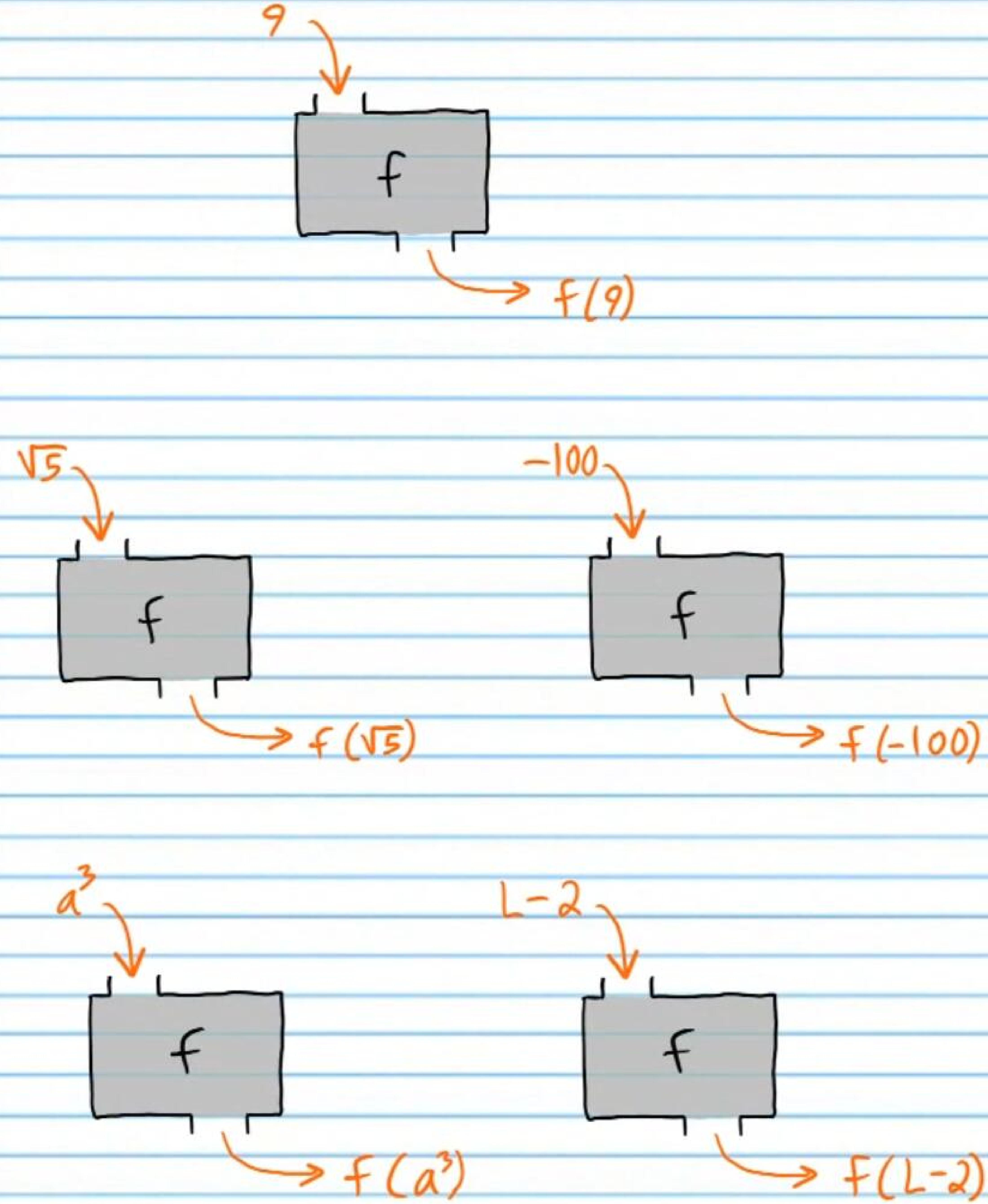



f(x) is pronounced "f of x."

Warning: Although this notation is most commonly used, it can be very confusing because the parentheses on f(x) DO NOT indicate multiplication.

# Using f(x) Notation

## Example V



① If $f(x) = 2x - 5$, find i) $f(3)$, ii) $f(-4)$, and iii) $f(0)$.

i) $f(3) = 2(3) - 5 = 6 - 5 = \boxed{1}$

ii) $f(-4) = 2(-4) - 5 = -8 - 5 = \boxed{-13}$

iii) $f(0) = 2(0) - 5 = 0 - 5 = \boxed{-5}$

② If $f(x) = 2^x - 9$, find i) $f(2)$, ii) $f(3)$, and iii) $f(-1)$.

i) $f(2) = 2^2 - 9 = 4 - 9 = \boxed{-5}$

ii) $f(3) = 2^3 - 9 = 8 - 9 = \boxed{-1}$

iii) $f(-1) = 2^{-1} - 9 = \frac{1}{2^1} - 9 = \frac{1}{2} - 9 = \boxed{-8\frac{1}{2}}$

③ If $f(x) = (x+1)^2$, find i) $f(-2)$, ii) $f(5)$, and iii) $f(-7)$.

i) $f(-2) = (-2+1)^2 = (-1)^2 = \boxed{1}$

ii) $f(5) = (5+1)^2 = 6^2 = \boxed{36}$

iii) $f(-7) = (-7+1)^2 = (-6)^2 = \boxed{36}$

④ If $f(x) = x^2 + 3x - \frac{1}{2}$, find i) $f(0)$, ii) $f(2)$, and iii) $f(1)$.

i) $f(0) = 0^2 + 3(0) - \frac{1}{2} = 0 + 0 - \frac{1}{2} = \boxed{-\frac{1}{2}}$

ii) $f(2) = 2^2 + 3(2) - \frac{1}{2} = 4 + 6 - \frac{1}{2} = \boxed{9\frac{1}{2}}$

iii) $f(1) = 1^2 + 3(1) - \frac{1}{2} = 1 + 3 - \frac{1}{2} = \boxed{3\frac{1}{2}}$

⑤ If $f(x) = \sqrt{x+7} + 3x^2$, find i) $f(-3)$, ii) $f(-6)$, and iii) $f(4)$.

i) $f(-3) = \sqrt{-3+7} + 3(-3)^2 = \sqrt{4} + 3(9) = 2 + 27$
$= \boxed{29}$

ii) $f(-6) = \sqrt{-6+7} + 3(-6)^2 = \sqrt{1} + 3(36) = 1 + 108$
$= \boxed{109}$

iii) $f(4) = \sqrt{4+7} + 3(4)^2 = \sqrt{11} + 3(16) = \boxed{\sqrt{11} + 48}$

⑥ If $f(x) = |\sqrt[3]{x}| + 2x - 4$, find i) $f(8)$, ii) $f(2)$, and iii) $f(1)$.

i) $f(8) = |\sqrt[3]{8}| + 2(8) - 4 = |2| + 16 - 4 = 2 + 16 - 4$
$= \boxed{14}$

ii) $f(2) = |\sqrt[3]{2}| + 2(2) - 4 = \sqrt[3]{2} + 4 - 4 = \boxed{\sqrt[3]{2}}$

iii) $f(1) = |\sqrt[3]{1}| + 2(1) - 4 = |1| + 2 - 4 = 1 + 2 - 4$
$= \boxed{-1}$

⑦ If $f(x)=\dfrac{\sqrt{x^2+4}}{7+\frac{2}{x}}$, find i) $f(1)$, ii) $f(2)$, and iii) $f(-2)$.

i) $f(1)=\dfrac{\sqrt{1^2+4}}{7+\frac{2}{1}}=\dfrac{\sqrt{1+4}}{7+2}=\boxed{\dfrac{\sqrt{5}}{9}}$

ii) $f(2)=\dfrac{\sqrt{2^2+4}}{7+\frac{2}{2}}=\dfrac{\sqrt{4+4}}{7+1}=\dfrac{\sqrt{8}}{8}=\dfrac{2\sqrt{2}}{8}=\boxed{\dfrac{\sqrt{2}}{4}}$

iii) $f(-2)=\dfrac{\sqrt{(-2)^2+4}}{7+\frac{2}{-2}}=\dfrac{\sqrt{4+4}}{7+(-1)}=\dfrac{\sqrt{8}}{6}=\dfrac{2\sqrt{2}}{6}=\boxed{\dfrac{\sqrt{2}}{3}}$

⑧ If $f(x)=\dfrac{10}{x}+2$, fill in the table below.

| $x$ | $f(x)$ |
|---|---|
| $-5$ | 0 |
| $-2$ | $-3$ |
| 4 | 4.5 |
| 0 | und |

i) $f(-5)=\dfrac{10}{-5}+2=-2+2=\boxed{0}$

ii) $f(-2)=\dfrac{10}{-2}+2=-5+2=\boxed{-3}$

iii) $f(4)=\dfrac{10}{4}+2=\dfrac{5}{2}+2=\boxed{4.5}$

iv) $f(0)=\dfrac{10}{0}+2=\boxed{\text{und}}$

⑨ If $f(x)=3-4|x|$, fill in the table below.

| $x$ | $-3$ | 0 | $\pi$ | $-0.6$ |
|---|---|---|---|---|
| $f(x)$ | $-9$ | 3 | $3-4\pi$ | 0.6 |

i) $f(-3)=3-4|-3|=3-4(3)=3-12=\boxed{-9}$

ii) $f(0)=3-4|0|=3-4(0)=\boxed{3}$

iii) $f(\pi)=3-4|\pi|=\boxed{3-4\pi}$

iv) $f(-0.6)=3-4|-0.6|=3-4(0.6)=3-2.4=\boxed{0.6}$

⑩ If $f(x)=3-\dfrac{8}{x^2}$, fill in the table below.

| $x$ | $f(x)$ |
|---|---|
| 2 | 1 |
| $-1$ | $-5$ |
| $-9$ | $235/81$ |

i) $f(2)=3-\dfrac{8}{2^2}=3-\dfrac{8}{4}=3-2=\boxed{1}$

ii) $f(-1)=3-\dfrac{8}{(-1)^2}=3-\dfrac{8}{1}=3-8=\boxed{-5}$

iii) $f(-9)=3-\dfrac{8}{(-9)^2}=3-\dfrac{8}{81}=\dfrac{243}{81}-\dfrac{8}{81}=\boxed{\dfrac{235}{81}}$

⑪ If $f(x)=3^{x+2}$, fill in the table below.

| $x$ | 0 | 1 | $-1$ | $-2$ |
|---|---|---|---|---|
| $f(x)$ | 9 | 27 | 3 | 1 |

i) $f(0)=3^{0+2}=3^2=\boxed{9}$

ii) $f(1)=3^{1+2}=3^3=\boxed{27}$

iii) $f(-1)=3^{-1+2}=3^1=\boxed{3}$

iv) $f(-2)=3^{-2+2}=3^0=\boxed{1}$

(12) If $f(x) = -8x^3$, find i) $f(a)$, ii) $f(b+2)$, and iii) $f(2/3)$.

i) $f(a) = \boxed{-8a^3}$

ii) $f(b+2) = \boxed{-8(b+2)^3}$

iii) $f\left(\frac{2}{3}\right) = -8\left(\frac{2}{3}\right)^3 = -8\left(\frac{8}{27}\right) = \boxed{-\frac{64}{27}}$

(13) If $f(x) = x^{\frac{3}{4}} + x$, find i) $f(y)$, ii) $f(c^2)$, and iii) $f(b^2+2b+6)$.

i) $f(y) = \boxed{y^{\frac{3}{4}} + y}$

ii) $f(c^2) = (c^2)^{3/4} + c^2 = c^{2 \cdot 3/4} + c^2 = c^{3/2} + c^2 = \boxed{(\sqrt{c})^3 + c^2}$

iii) $f(b^2+2b+6) = (b^2+2b+6)^{\frac{3}{4}} + b^2+2b+6$

$= \boxed{\left(\sqrt[4]{b^2+2b+6}\right)^3 + b^2+2b+6}$

(14) If $f(x) = \frac{x+2}{x-2}$, find i) $f(D)$, ii) $f(L^{10}-2)$, and iii) $f(3.1)$.

i) $f(D) = \boxed{\frac{D+2}{D-2}}$

ii) $f(L^{10}-2) = \frac{L^{10}-2+2}{L^{10}-2-2} = \boxed{\frac{L^{10}}{L^{10}-4}}$

iii) $f(3.1) = \frac{3.1+2}{3.1-2} = \frac{5.1}{1.1} = \boxed{4.\overline{63}}$

(15) If $f(x) = 4^x + \frac{6}{x}$, find i) $f(-\frac{1}{2})$, ii) $f(3b)$, and iii) $f(10^a+3a)$.

i) $f\left(-\frac{1}{2}\right) = 4^{-\frac{1}{2}} + \frac{6}{-\frac{1}{2}} = \frac{1}{4^{\frac{1}{2}}} + 6 \div \left(-\frac{1}{2}\right) = \frac{1}{\sqrt{4}} + 6\left(-\frac{2}{1}\right) = \frac{1}{2} - 12$

$= \boxed{-11\frac{1}{2}}$

ii) $f(3b) = 4^{3b} + \frac{6}{3b} = \boxed{4^{3b} + \frac{2}{b}}$

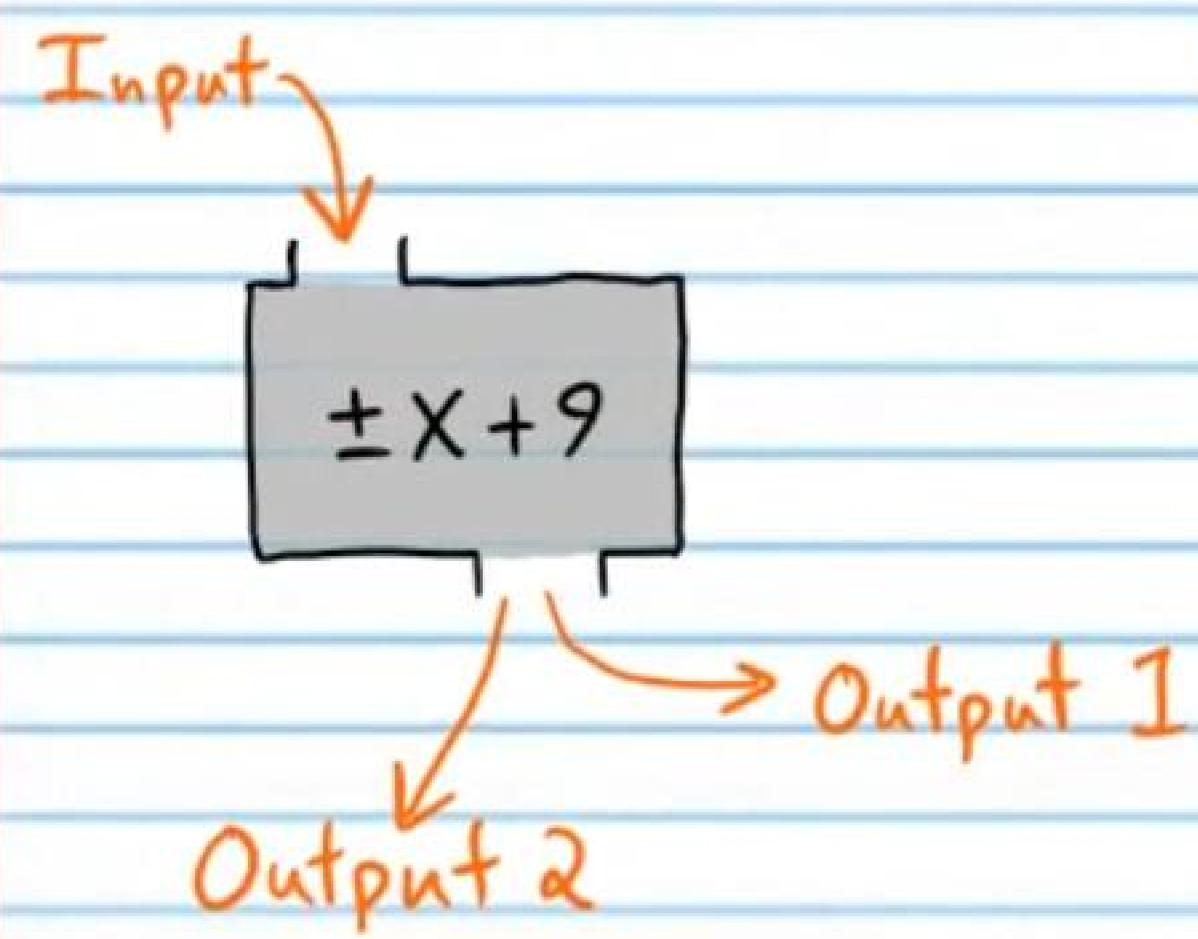
iii) $f(10^a+3a) = \boxed{4^{10^a+3a} + \frac{6}{10^a+3a}}$

## A Neccessary Characteristic of Functions

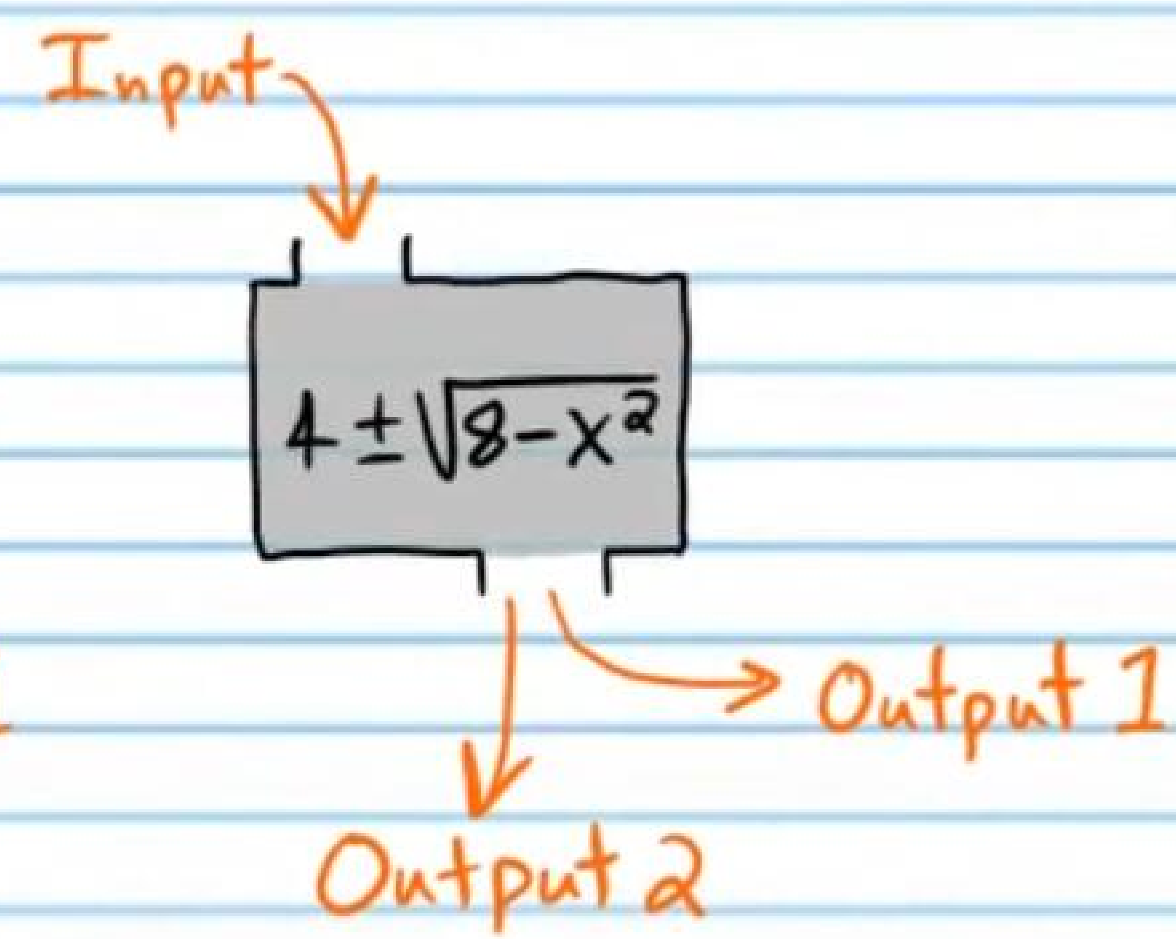
In order for an expression to qualify as a function, every input must produce ONLY ONE output.

These are not functions because the expressions have ($\pm$) symbols on them, which indicate that for some inputs, there are two resulting outputs.

Example VI




Example VII



Let's try some inputs.

5 → $\pm 5 + 9$

$5+9=\boxed{14}$

$-5+9=\boxed{4}$

$-1$ → $4 \pm \sqrt{8-(-1)^2}$

$4+\sqrt{8-(-1)^2}$

$4+\sqrt{8-1}$

$\boxed{4+\sqrt{7}}$

$4-\sqrt{8-(-1)^2}$

$4-\sqrt{8-1}$

$\boxed{4-\sqrt{7}}$

In the tables below, x represents the inputs and y represents the outputs. Circle the tables that could not represent functions.

16

| x | y |
|---|---|
| 3 | -2 |
| 8 | 10 |
| 0 | 4 |
| -8 | -7 |
| 1 | 5 |
| -2 | 0 |

| x | y |
|---|---|
| 0 | 1 |
| 3 | 5 |
| -7 | -8 |
| 4 | 3 |
| 0 | 9 |
| 1 | 6 |

| x | y |
|---|---|
| 36 | 40 |
| 7 | 38 |
| 20 | 26 |
| 85 | 20 |
| -100 | 10 |
| 67 | 9 |

| x | y |
|---|---|
| 7 | 16 |
| 7 | 5 |
| 8 | 2 |
| -2 | 4 |
| 9 | 11 |
| 10 | 3 |

17

| x | y |
|---|---|
| -4 | 2 |
| 8 | -3 |
| 10 | 12 |
| 13 | 17 |
| 10 | 1/3 |
| -4 | 1 |

| x | y |
|---|---|
| 0 | 13 |
| 2 | 3 |
| -3 | 17 |
| 6 | 8 |
| -3 | 10 |
| 6 | -5 |

| x | y |
|---|---|
| -7 | 2 |
| 8 | 4 |
| 10 | 5 |
| -9 | 3 |
| -2 | -1 |
| 0 | 0 |

| x | y |
|---|---|
| 200 | 500 |
| 300 | 600 |
| 400 | 700 |
| 500 | 800 |
| 600 | 900 |
| 700 | 1000 |

18

| x | y |
|---|---|
| 1 | 2 |
| 5 | 8 |
| 2 | 13 |
| 4 | 0 |
| 9 | 1 |
| -1 | 7 |

| x | y |
|---|---|
| -7 | 8 |
| 9 | 2 |
| -6 | -3 |
| -7 | 10 |
| 2 | 0 |
| 3 | 5 |

| x | y |
|---|---|
| 14 | -17 |
| 8 | -15 |
| -9 | 0 |
| -2 | 3 |
| 20 | -16 |
| 31 | -11 |

| x | y |
|---|---|
| 2 | -3 |
| 4 | -2 |
| 9 | 0 |
| 9 | 1 |
| 2.3 | 8 |
| -0.7 | 0.9 |

19

| x | y |
|---|---|
| -1 | -30 |
| 0 | -20 |
| 1 | -10 |
| 8 | -3 |
| 9 | 5 |
| 12 | -9 |

| x | y |
|---|---|
| 1 | 10 |
| 2 | 12 |
| 3 | 14 |
| 4 | 16 |
| 5 | 18 |
| 6 | 20 |

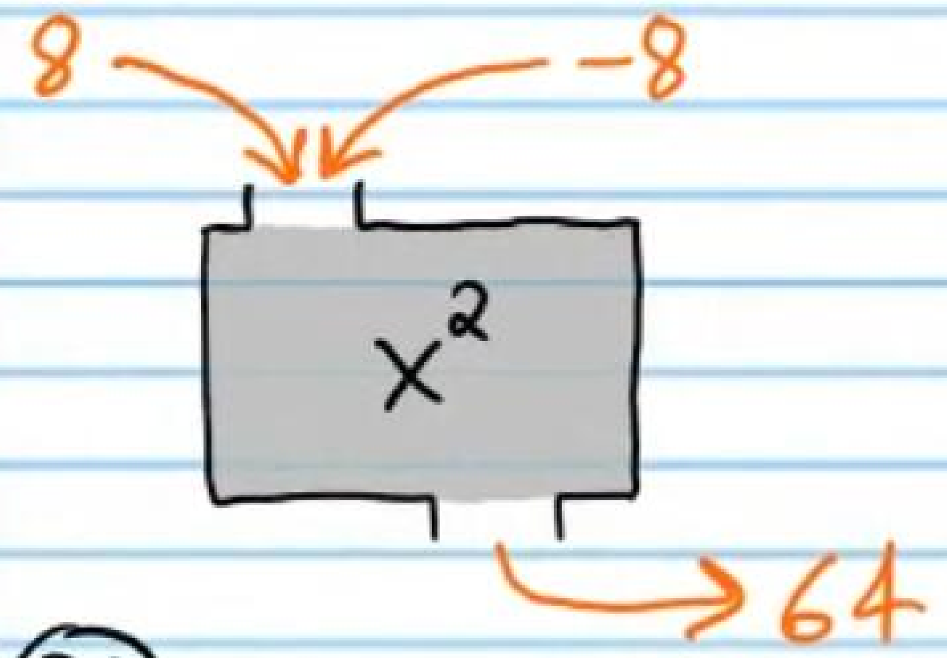
| x | y |
|---|---|
| 42 | 45 |
| 56 | 23 |
| 68 | 43 |
| 20 | 40 |
| 56 | 21 |
| 20 | 39 |

| x | y |
|---|---|
| 3 | 10 |
| 8 | 9 |
| 3 | 8 |
| -3 | 7 |
| -8 | 6 |
| 8 | 5 |

Warning: If different inputs produce the same output in an expression (Example VIII), then this does not disqualify the expression from being a function.

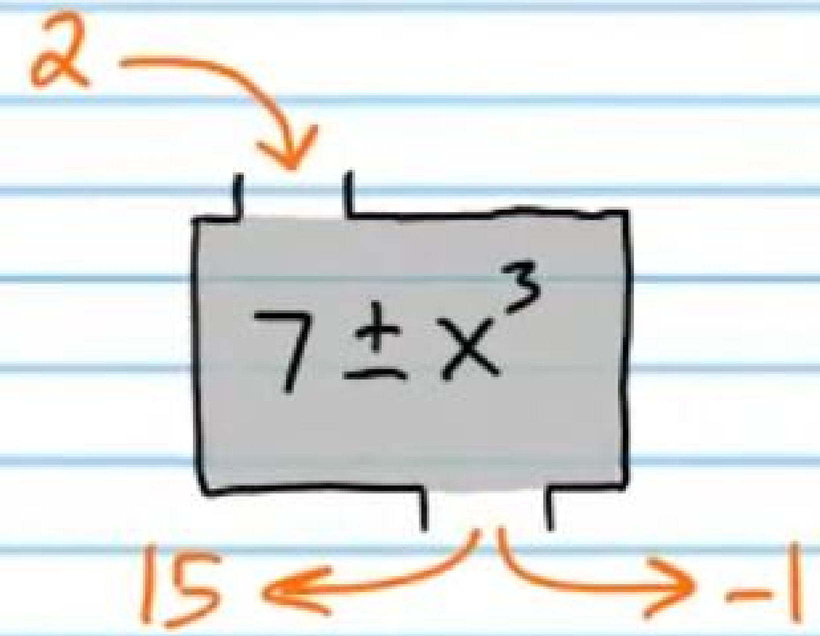
Example VIII

★ Can be a function.




Example IX

★ Cannot be a function.




(20)

| X | 6 | 0 | −2 | 3 | 10 | 19 |
|---|---|---|---|---|---|---|
| Y | 3 | 8 | 9 | −2 | 5 | 1 |

| X | −1 | 2 | 4 | −5 | 6 | −5 |
|---|---|---|---|---|---|---|
| Y | 3 | 8 | 8 | 1 | 0 | −1 |

| X | 3 | 5 | 6 | 0 | −8 | 10 |
|---|---|---|---|---|---|---|
| Y | −4 | −1 | 2 | 2 | 3 | 5 |

| X | −3 | −8 | 9 | 10 | 12 | 13 |
|---|---|---|---|---|---|---|
| Y | 1 | 1 | 1 | 2 | 3 | 1 |

(21)

| X | 10 | −7 | 10 | 9 | 10 | 3 |
|---|---|---|---|---|---|---|
| Y | 8 | 4 | 3 | 1 | 7 | 6 |

| X | 6 | 2 | −1 | 17 | 11 | 12 |
|---|---|---|---|---|---|---|
| Y | 4 | 3 | 3 | 0 | 0 | 4 |

| X | Y |
|---|---|
| −1 | 2 |
| −8 | 0 |
| −5 | 2 |
| 3 | 2 |
| 9 | 1 |
| 10 | 7 |

| X | Y |
|---|---|
| 7 | 8 |
| 6 | 8 |
| 5 | 8 |
| 4 | 8 |
| 3 | 8 |
| 2 | 8 |

(22)

| X | 70 | 40 | 20 | −20 | 41 | 71 |
|---|---|---|---|---|---|---|
| Y | 3 | 8 | 5 | 4 | 2 | 6 |

| X | 7 | 10 | 6 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|
| Y | 5 | 3 | 5 | 2 | 1 | 0 |

| X | 0 | 1 | −1 | 2 | 0 | −2 |
|---|---|---|---|---|---|---|
| Y | 6 | 3 | 4 | 17 | 17 | 18 |

| X | 3 | 6 | −2 | 4 | 5 | 7 |
|---|---|---|---|---|---|---|
| Y | 9 | 9 | 0 | 3 | 6 | 1 |

(23)

| X | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|
| Y | 10 | 10 | 10 | 10 | 7 | 6 |

| X | −1 | −5 | 0 | −5 | 4 | 3 |
|---|---|---|---|---|---|---|
| Y | 36 | 35 | 36 | 36 | 36 | 36 |

| X | Y |
|---|---|
| 7 | 6 |
| 6 | 7 |
| 5 | 5 |
| 4 | 5 |
| 3 | 1 |
| 3 | 6 |

| X | Y |
|---|---|
| 8 | 19 |
| 12 | 3 |
| 9 | 2 |
| 8 | 4 |
| 9 | 0 |
| 12 | 19 |

(24)

| X | 20 | 3 | 7 | −1 | 0 | 7 |
|---|---|---|---|---|---|---|
| Y | 0 | 8 | −4 | 2 | 18 | −4 |

| X | 10 | 5 | 4 | 2 | 4 | 10 |
|---|---|---|---|---|---|---|
| Y | −10 | 8 | 2 | 1 | 3 | −10 |

| X | Y |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |
| 4 | 4 |
| 5 | 5 |
| 6 | 6 |

| X | Y |
|---|---|
| −3 | 9 |
| −3 | 9 |
| −2 | 0 |
| −1 | −1 |
| 0 | 0 |
| 1 | 2 |

(25)

| x | y |
|---|---|
| -7 | -5 |
| 2 | -5 |
| 3 | -5 |
| 4 | -5 |
| 0 | 4 |
| -7 | -5 |

| x | y |
|---|---|
| 4 | 3 |
| 8 | 3 |
| 9 | 0 |
| 10 | -1 |
| 11 | -2 |
| 11 | -2 |

| x | y |
|---|---|
| 8 | 0 |
| 8 | 0 |
| 8 | 0 |
| 4 | 7 |
| 3 | 6 |
| -2 | 5 |

| x | y |
|---|---|
| 5 | 2 |
| 6 | 3 |
| 5 | 2 |
| 6 | 8 |
| 4 | -2 |
| 2 | 0 |

(26)

| X | 0 | 1 | 17 | 16 | 16 | 18 |
|---|---|---|---|---|---|---|
| Y | 8 | 3 | 10 | 9 | 9 | 11 |

| X | 3 | 2 | 0 | -3 | -5 | -3 |
|---|---|---|---|---|---|---|
| Y | 4 | 4 | 8 | 7 | 6 | 7 |

| X | -9 | -3 | -3 | -3 | 6 | 7 |
|---|---|---|---|---|---|---|
| Y | 10 | 9 | 8 | 7 | 6 | 5 |

| X | 0 | 1 | 2 | -6 | -6 | -6 |
|---|---|---|---|---|---|---|
| Y | 8 | 8 | 10 | 14 | 14 | 15 |

(27)

| X | 14 | 12 | 10 | 10 | 14 | 8 |
|---|---|---|---|---|---|---|
| Y | 7 | 5 | 7 | 7 | 7 | 5 |

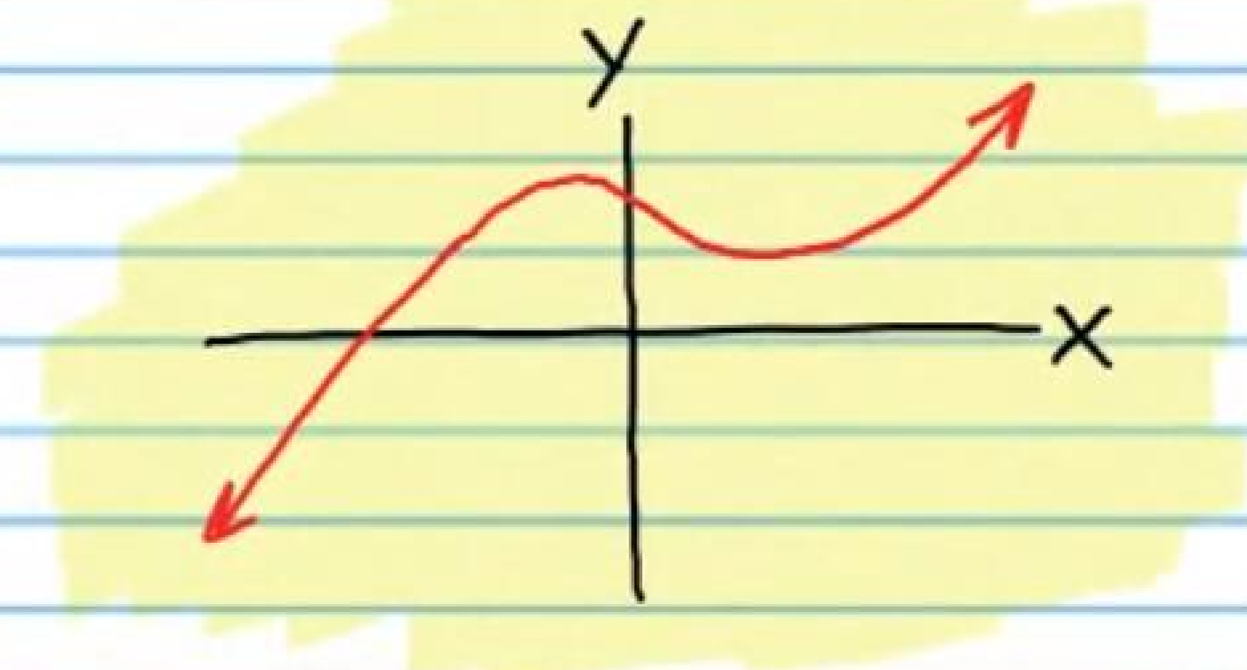
| X | -6 | -2 | 4 | 3 | 3 | 4 |
|---|---|---|---|---|---|---|
| Y | 3 | 0 | 1 | -8 | -9 | 1 |

| x | y |
|---|---|
| 0 | 16 |
| 5 | 13 |
| 19 | 8 |
| 0 | 16 |
| 10 | 2 |
| 11 | 3 |

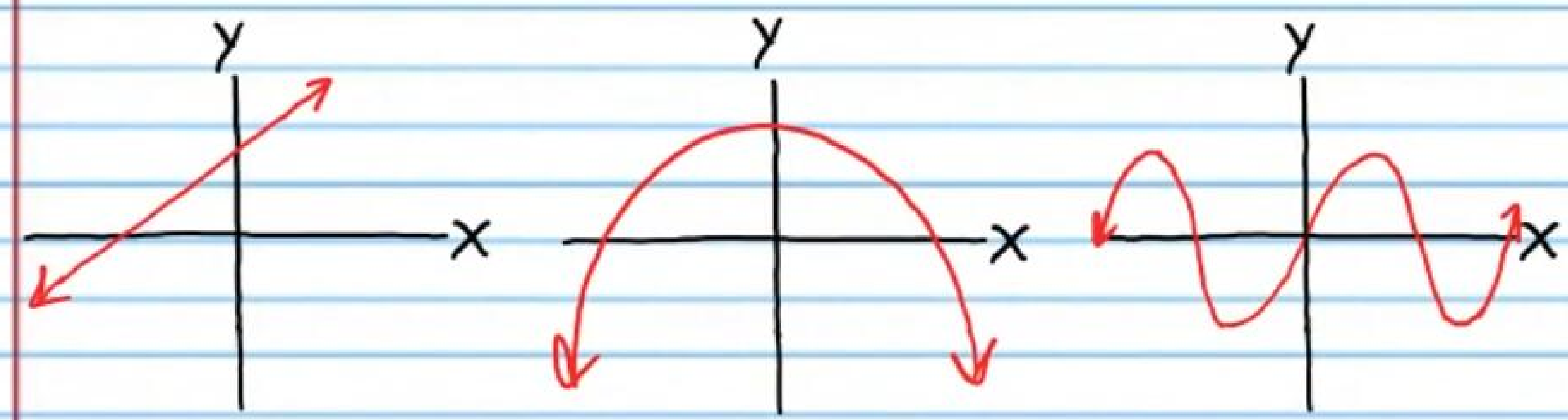
| x | y |
|---|---|
| 8 | 15 |
| 8 | 15 |
| -5 | 3 |
| -2 | 0 |
| 0 | 0 |
| -5 | 100 |

# Three Ways to Represent a Function

I. Equation
II. Table
III. Graph



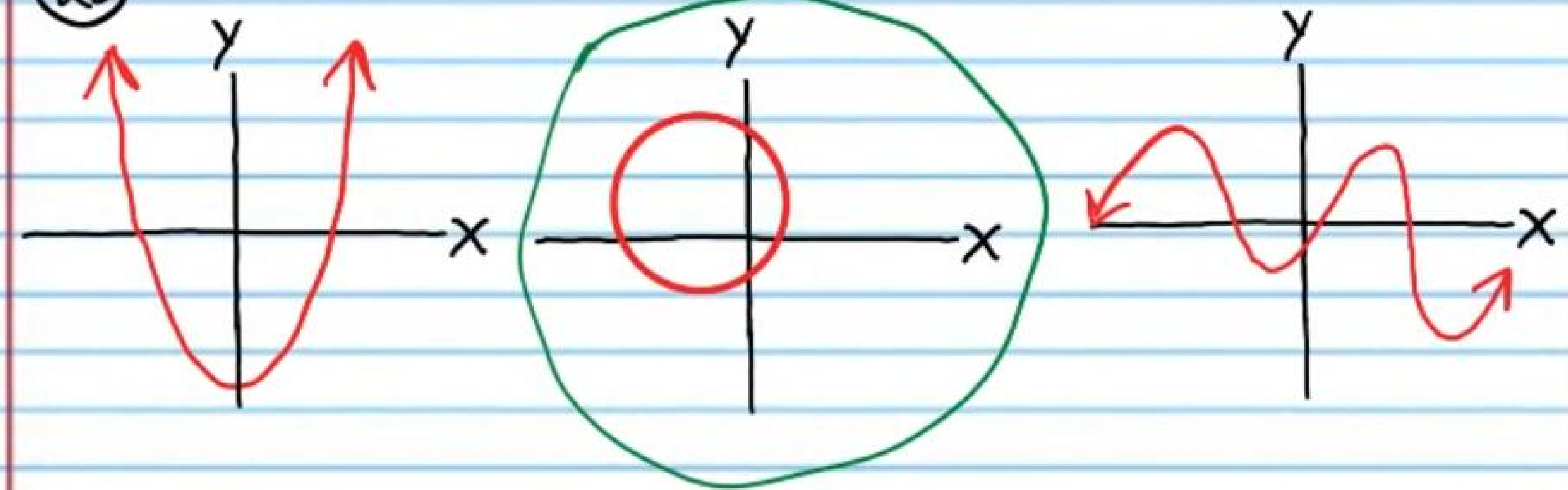

## Examples of Graphs of Functions



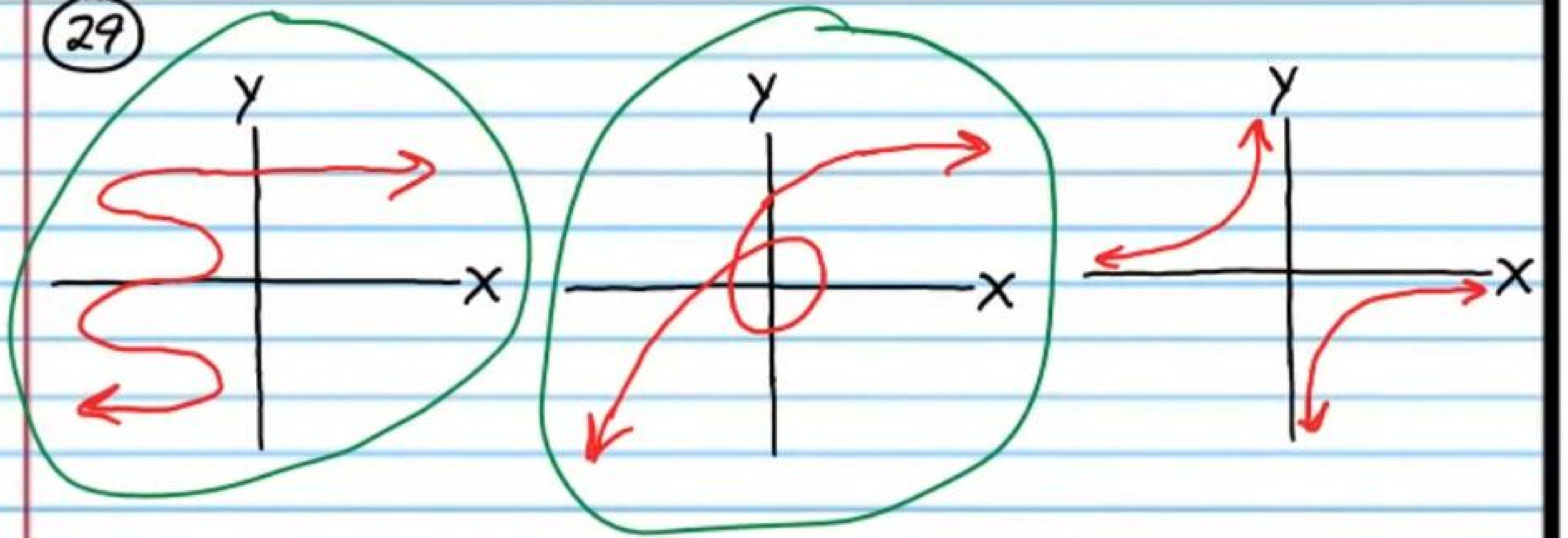
The Vertical Line Test: If any vertical line touches a graph more than once, then the graph cannot be a function.

Circle the relations below that cannot be functions.

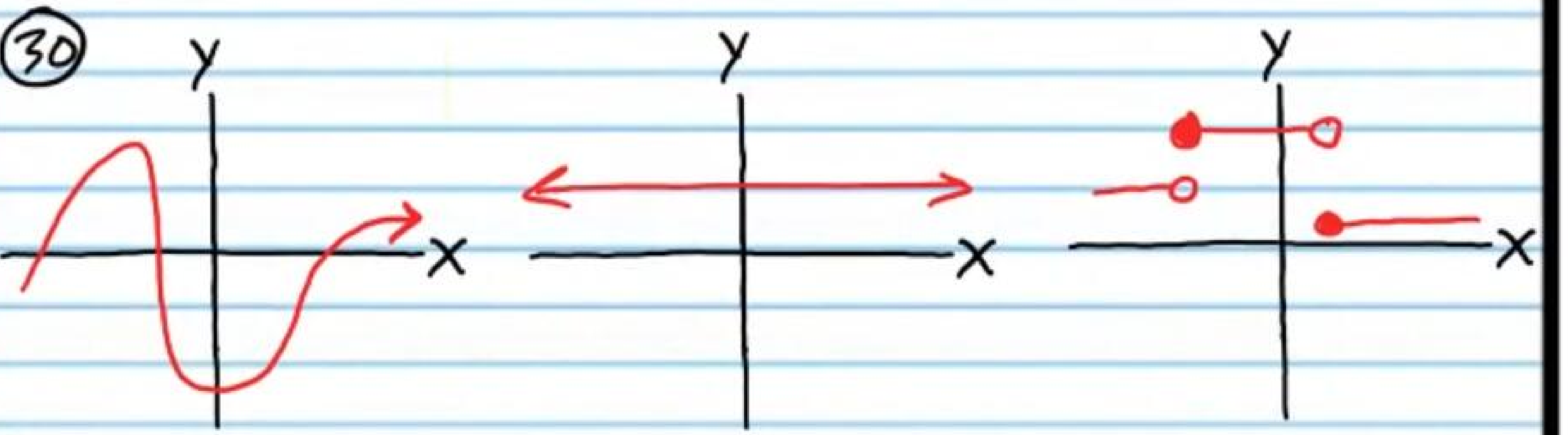
(28)



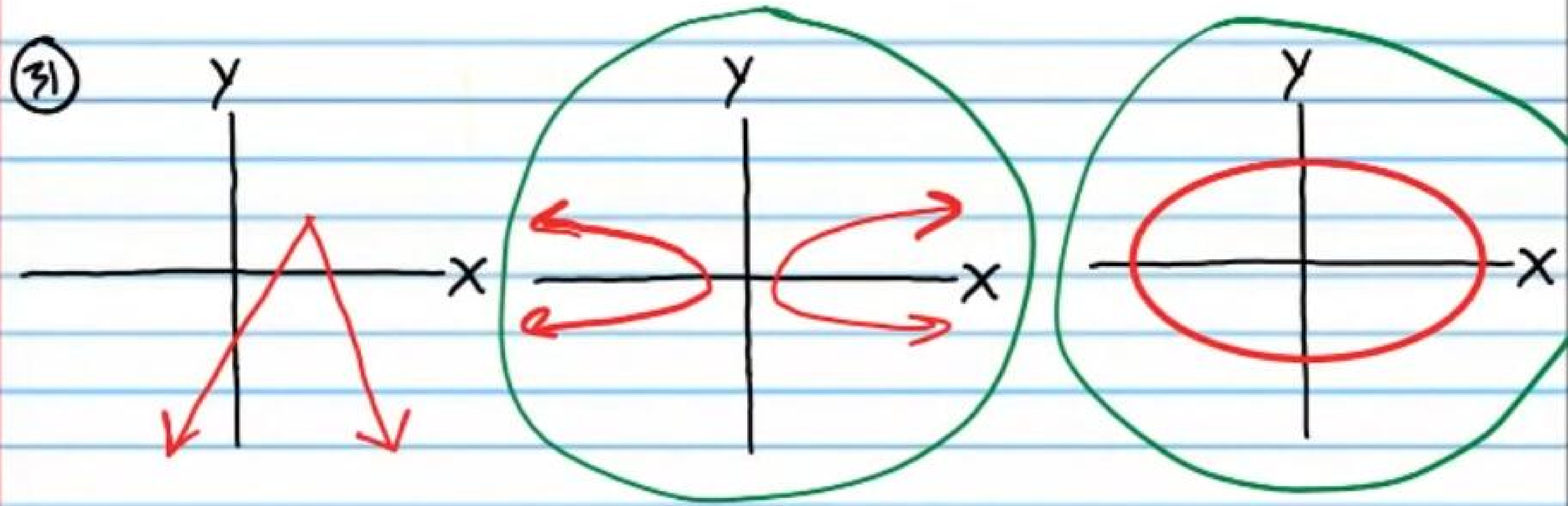

(29)



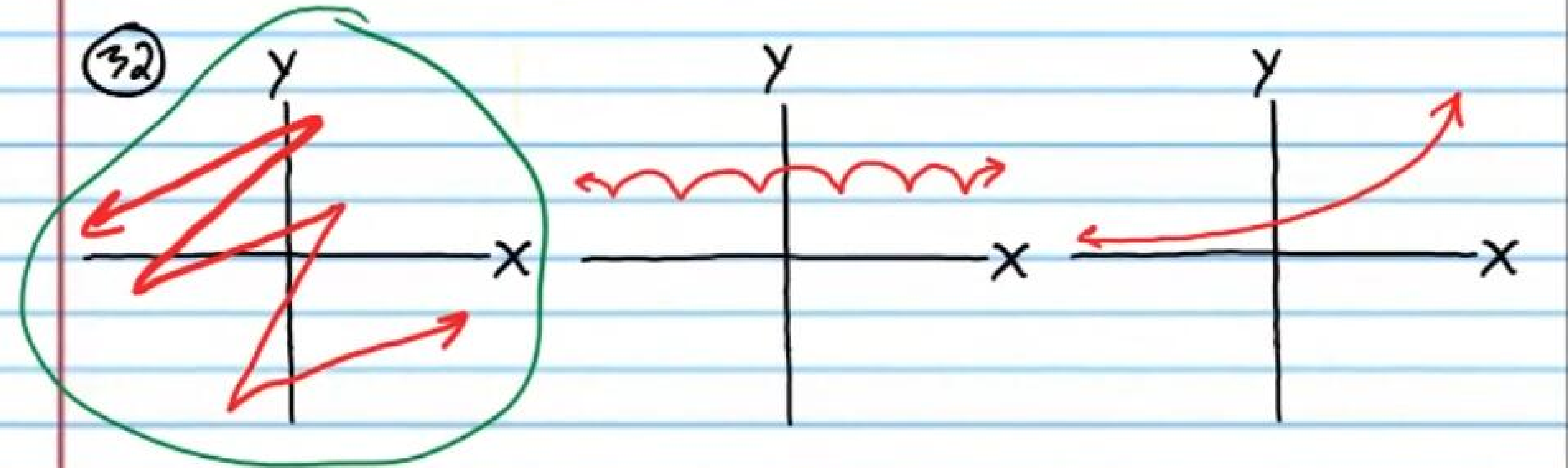

(30)




(31)




(32)




(33)